

Identifying and Predicting Successful Markets on Polymarket

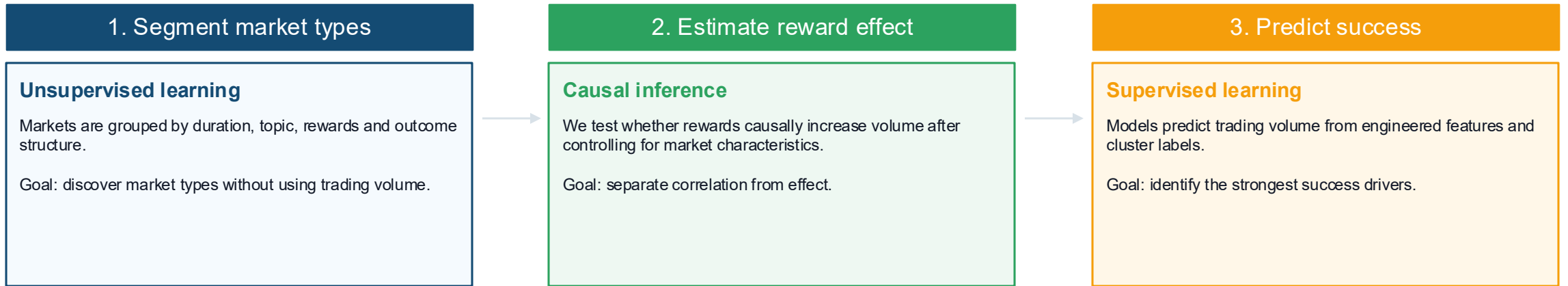
Can machine learning identify distinct types of prediction markets on Polymarket, and do these market characteristics help predict market success measured by trading volume? Additionally, we investigate whether market rewards causally affect market success.

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Motivation & Economic Relevance

Prediction markets aggregate beliefs into prices. We ask why some Polymarket markets attract meaningful trading volume while others stay inactive.

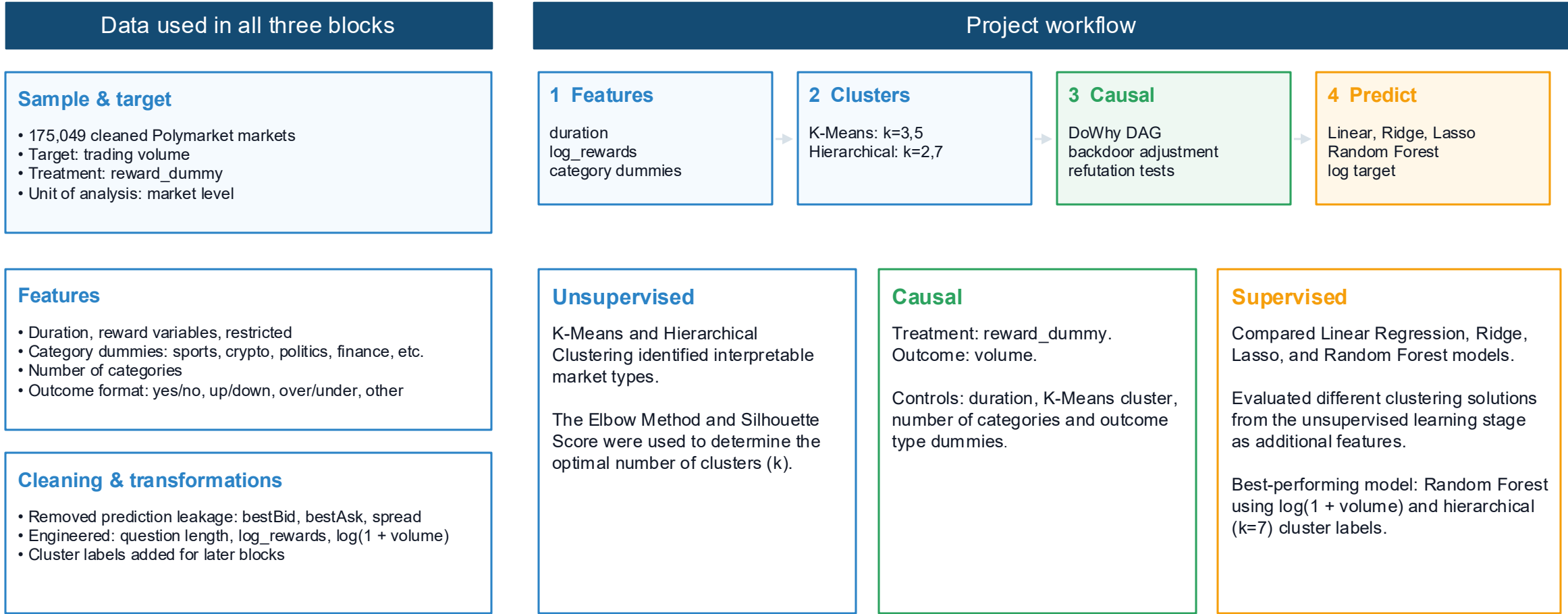


Economic relevance

If market success can be explained and predicted, platforms can prioritize promising markets and allocate rewards where they create additional participation.

Core logic: segmentation → reward effect → prediction of trading volume.

Data & Methodology



Results

SAMPLE

175k

cleaned markets

REWARD ATE

+316k to +343k

volume effect

BEST R²

0.4805

RF + log target

VOLUME MAX

1.53B

extreme skew

TOP DRIVER

duration

feature importance

Segmentation

K-Means k=5 market types

- Long-term political and sports markets
- Uncategorized short-term markets
- High-reward general event markets
- Short crypto price markets
- Short sports betting markets

Interpretation

Cluster labels summarize market structure and become features/confounders in later models.

Causal effect of rewards

Rewards increase volume after adjustment

Method	ATE on trading volume
Propensity stratification	+316,192
Linear regression	+343,360
Avg	329,776

Robustness

Random common cause, placebo treatment and data subset refutations passed.

Prediction

Model Comparison (Hierarchical k=7)

Model	R ² Score
Linear Regression	0.383
Ridge	0.383
Lasso	0.383
RF (raw)	-0.064
RF (log)	0.480

Key result

Random Forest performs well only after log(1 + volume), explaining about 48% of variation (R² = 0.48).

Main result: rewards have a positive causal effect and volume is predictable once the skewed target is transformed.

Limitations & Synthesis

Main limitations

1. Observational design

Rewards were not randomly assigned. DoWhy adjusts for observed confounders, but unobserved platform choices or promotion effects may remain.

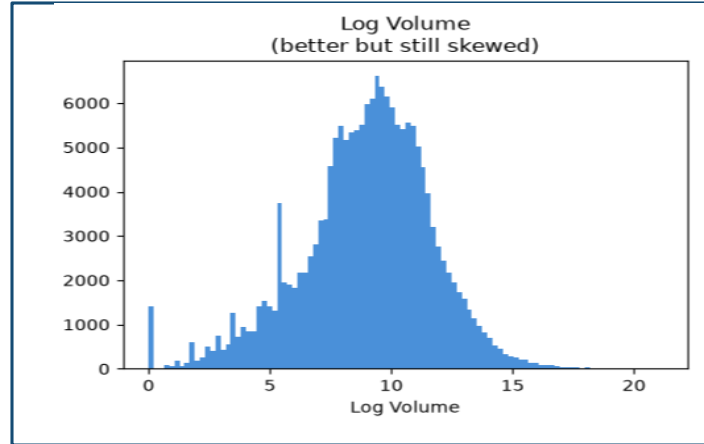
2. Market-level features only

The analysis does not include trader networks, time-varying prices, order-book dynamics or external news shocks.

3. Cluster choice

k was selected using the Elbow Method, silhouette score, dendrograms and interpretability.

Why prediction was difficult



Modeling implication

Raw trading volume is dominated by a few extreme markets. Log transformation was the central improvement.

Final synthesis

Finding 1

Markets are heterogeneous: duration, topic and outcome format create distinct market types.

Finding 2

Rewards likely increase participation: controlled ATE is around +316k to +343k volume.

Finding 3

A log-target Random Forest predicts a meaningful share of volume variation ($R^2 \approx 0.48$).

Final takeaway

Machine learning identifies market types and predicts success; causal inference suggests rewards are not only correlated with volume but help create additional market activity.

References

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